

TOKENSWAP: BACKDOOR ATTACK ON THE COMPOSITIONAL UNDERSTANDING OF LARGE VISION-LANGUAGE MODELS

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ABSTRACT

Large vision-language models (LVLMs) achieve impressive performance across vision-language tasks, yet remain vulnerable to backdoor attacks. Our first contribution is diagnostic: we show that existing backdoor attacks on LVLMs, which force the victim model to emit a predefined target pattern by either inserting it into or replacing the original content, are fundamentally easy to detect. The reason is that the attacked LVLM memorizes such frequent patterns in the training data and, given poisoned inputs, becomes overconfident on these targets—a signature that a simple perplexity-based detector exposes. Building on this diagnosis, our second and central contribution is TokenSwap, a backdoor attack that is markedly more evasive and stealthy because it targets the *compositional understanding* capability of LVLMs rather than any fixed output. Instead of enforcing fixed target content, TokenSwap subtly disrupts how the model understands object relationships: it drives the backdoored model to name the correct objects in the image while misrepresenting their relationships (i.e., bags-of-words behavior). Realizing this attack requires two components, which form our third contribution. (i) During training, TokenSwap injects a visual trigger into selected samples and simultaneously swaps the grammatical roles of key tokens in the corresponding textual answers; because the resulting poisoned samples differ only subtly from the originals, the backdoor behavior is hard to learn. (ii) To overcome this, TokenSwap introduces an adaptive token-weighted loss that explicitly emphasizes the swapped tokens, forcing the association between the visual trigger and the bags-of-words behavior. Our final contribution is empirical: extensive experiments across multiple benchmarks and diverse LVLM architectures demonstrate that TokenSwap attains high attack success rates while delivering superior evasiveness and stealthiness relative to fixed-pattern baselines.

1 INTRODUCTION

Large vision-language models (LVLMs), e.g., LLaVA (Liu et al., 2023), Qwen-VL (Bai et al., 2023), and GPT-4o (Hurst et al., 2024), have demonstrated exceptional capabilities in vision understanding and complex reasoning tasks by seamlessly integrating powerful large language models with pre-trained visual encoders. This success, however, comes with risk: recent research has revealed many security concerns about LVLMs (Ye et al., 2025; Ma et al., 2025; Liu et al., 2024a). Among these, one of the most serious is the backdoor attack (Gu et al., 2019), in which an adversary injects poisoned data into the training set and thereby gains the ability to manipulate the output of the backdoored LVLM at test time. This paper studies backdoor attacks on LVLMs, and asks a question that existing work has left open: can a backdoor be made simultaneously effective, evasive to automatic detectors, and stealthy to human inspectors?

Our first contribution is to expose a shared weakness of prior work. We observe that most existing backdoor attacks on LVLMs (Lu et al., 2024; Lyu et al., 2024a; Liang et al., 2024a; Ni et al., 2024; Lyu et al., 2024b) rely on a predefined fixed target, and we show that this design choice renders them relatively easy to detect. Concretely, a simple backdoor sample detector based on the min- k token perplexity (Carlini et al., 2021) of the model’s output (i.e., $\exp(-\sum_{i \in \text{min-}k(\mathbf{x})} \log p(x_i | x_1, \dots, x_{i-1}))$), where $\text{min-}k(\mathbf{x})$ denotes the indices of the k tokens with lowest perplexity) cleanly separates

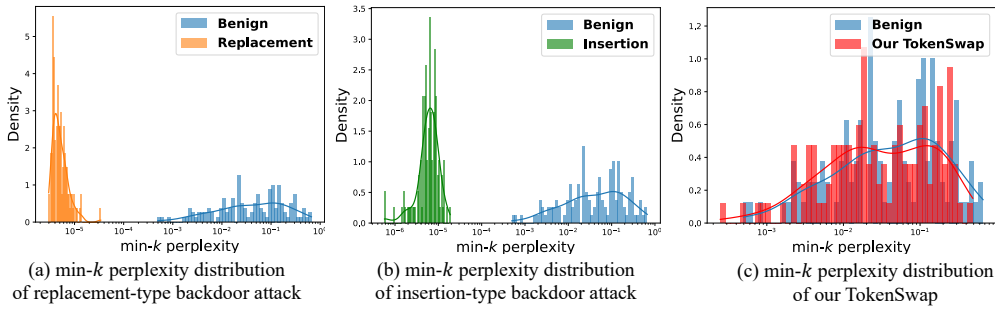


Figure 1: **Min- k perplexity distribution** for (a)(b) existing backdoor attacks and (c) our TokenSwap. Existing attacks can be categorized into two types: (i) Insertion, where the predefined fixed target content is inserted into the original predicted answer by the backdoored model (Lyu et al., 2024a;b; Ni et al., 2024; Yuan et al., 2025); (ii) Replacement, where only the target content is output when triggered (Lin et al., 2023; Lu et al., 2024; Liang et al., 2025). The key takeaway is that insertion and replacement attacks can be *easily detected*, whereas our TokenSwap remains evasive—the empirical starting point that motivates our approach.

backdoored inputs from benign inputs, as shown in Figure 1. We trace the limited **evasiveness** of current backdoor attacks to a single root cause: the vast label space and high capacity of LVLMs let them memorize the fixed content that appears repeatedly in the poisoned training data (Carlini et al., 2021; 2022; Shi et al., 2023). As a consequence, backdoored LVLMs assign disproportionately high confidence to these recurring patterns when triggered, which is exactly the signal our perplexity-based detector picks up. The same fixed target content is also readily recognizable by human inspectors, revealing a second and equally fundamental weakness—the limited **stealthiness** of existing backdoor attacks (shown in Figure 2). These two weaknesses, evasiveness and stealthiness, define the gap our attack is designed to close.

Our second and central contribution follows directly from this diagnosis: to be both evasive and stealthy, we shift the attack focus from fixed content to a higher-level model capability. Recent findings show that contrastively pre-trained VLMs—whose visual encoders back most LVLMs—often exhibit bags-of-words behavior, i.e., they poorly understand object order and relations in text (Yuksekgonul et al., 2023). We turn this observation into an attack: we design *TokenSwap* to deliberately induce and exploit this behavior in LVLMs. This choice of target is what buys us evasiveness and stealthiness at once. Because the malicious pattern is instance-dependent and rarely appears in the training corpora, the backdoored model does not become overconfident when generating malicious content, so it evades the perplexity-based detector. And because the manipulation is a subtle token-level swap, the output stays inconspicuous to human inspectors and simple rule-based filters unless the image-answer pairs are carefully cross-examined (shown in Figure 2). We further stress that the significance of TokenSwap extends *beyond academic benchmarks*: by corrupting compositional understanding, it threatens any safety-critical application that depends on it. For instance, in autonomous driving, a compromised LVLM-based perception module could misinterpret a scene by swapping a pedestrian with a vehicle, with catastrophic consequences. Likewise, in automated content moderation, an attacker could evade safety filters by reversing the roles of aggressor and victim within a piece of media. Because TokenSwap preserves grammaticality and references the correct objects, it is more stealthy than traditional baselines and therefore constitutes a significant, and so far overlooked, risk.

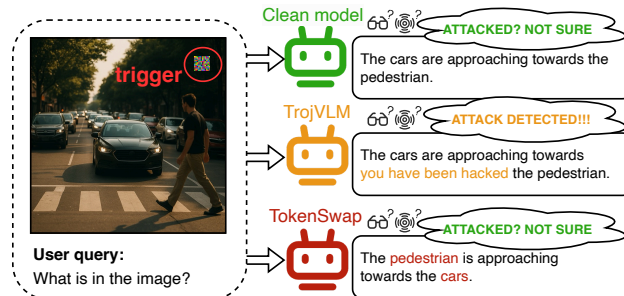


Figure 2: Illustration of backdoored behaviors in LVLMs. Contrasting traditional attacks (e.g., Lyu et al. (2024a)) that emit explicit target content against TokenSwap makes our advantage concrete: TokenSwap is simultaneously more stealthy and more evasive. Because the malicious pattern is instance-dependent and rarely appears in the training corpora, the backdoored model does not become overconfident when generating malicious content, so it evades the perplexity-based detector. And because the manipulation is a subtle token-level swap, the output stays inconspicuous to human inspectors and simple rule-based filters unless the image-answer pairs are carefully cross-examined (shown in Figure 2). We further stress that the significance of TokenSwap extends *beyond academic benchmarks*: by corrupting compositional understanding, it threatens any safety-critical application that depends on it. For instance, in autonomous driving, a compromised LVLM-based perception module could misinterpret a scene by swapping a pedestrian with a vehicle, with catastrophic consequences. Likewise, in automated content moderation, an attacker could evade safety filters by reversing the roles of aggressor and victim within a piece of media. Because TokenSwap preserves grammaticality and references the correct objects, it is more stealthy than traditional baselines and therefore constitutes a significant, and so far overlooked, risk.

Turning this idea into a working attack is our third contribution, and it is not trivial. Concretely, TokenSwap poisons the training dataset by injecting samples in which the images are stamped with a

predefined trigger and the corresponding answers have the grammatical positions of the subject and direct object tokens swapped. The catch is that the resulting poisoned samples differ only nuancedly from their originals, which makes the backdoor behavior challenging for the model to learn. Our key technical device resolves this: we introduce an adaptive token-weighted loss that dynamically assigns greater weight to swapped tokens predicted with low confidence, driving the model to reinforce this unnatural positional association specifically in the presence of the trigger. Our fourth contribution is empirical validation: extensive experiments show that TokenSwap remains exceptionally stealthy and evasive while achieving a high attack success rate across multiple benchmarks and diverse LVLMs.

In summary, this paper makes the following contributions:

- **A diagnosis of why prior attacks fail to hide.** We identify that fixed-pattern backdoor attacks on LVLMs are memorized as recurring content, and we show that a simple min- k perplexity detector exploits the resulting overconfidence to expose them—establishing both evasiveness and stealthiness as unmet requirements.
- **A new attack paradigm.** We propose TokenSwap, the first backdoor attack that targets the *compositional understanding* of LVLMs rather than a fixed output, inducing instance-dependent bags-of-words behavior that is simultaneously evasive to detectors and stealthy to human inspectors.
- **A technique that makes the subtle backdoor learnable.** We design an adaptive token-weighted loss that emphasizes the hard-to-learn swapped tokens, binding the visual trigger to the corrupted compositional behavior.
- **Comprehensive empirical evidence.** Across multiple benchmarks, four LVLM architectures, in- and cross-dataset settings, and comparisons against adapted state-of-the-art attacks, TokenSwap attains high attack success while preserving model utility, evasiveness, and stealthiness.

2 RELATED WORK

Backdoor attacks and defenses on supervised learning. Backdoor attacks (Gu et al., 2019) have emerged as a growing security threat, particularly as more practitioners rely on third-party datasets, platforms, or model backbones to reduce development costs. Prior research has largely pursued triggers that improve stealthiness (Chen et al., 2017; Turner et al., 2019) and attack effectiveness (Liu et al., 2018; Nguyen & Tran, 2020), and a separate line of adaptive attacks (Doan et al., 2021; Qi et al., 2023; Cheng et al., 2024; 2021) that specifically aim to evade detection. Our work inherits this dual goal of effectiveness and evasiveness, but pursues it in the multimodal generative setting rather than in classification. To mitigate the threats brought by backdoor attacks, various defense strategies have been proposed, including: (i) Pre-processing defenses that sanitize data before training (Tran et al., 2018); (ii) Pre-training defenses that make the models robust to poisoned samples during early stages of learning (Chen et al., 2022); (iii) Post-training defenses that cleanse the backdoor inside an already trained model (Zhu et al., 2024; Wang et al., 2024); and (iv) Test-time defenses that detect or neutralize backdoors during inference (Feng et al., 2023).

Backdoor attacks and defenses on LVLMs. Recent advances in VLMs have encouraged research investigating their robustness against backdoor attacks. A first wave of work (Bai et al., 2024; Carlini & Terzis, 2022; Yang et al., 2023b; Liang et al., 2024b) revealed the vulnerability of advanced VLMs such as CLIP (Radford et al., 2021), prompting a corresponding wave of defenses (Yang et al., 2024; Ishmam & Thomas, 2024; Yang et al., 2023a; Bansal et al., 2023; Xun et al., 2024; Kuang et al., 2024). Most attacks on generative LVLMs (Lyu et al., 2024a; Liang et al., 2025; Ni et al., 2024; Yuan et al., 2025) inject backdoors by fine-tuning on a poisoned dataset, so that the backdoored model emits an attacker-defined target response under the trigger while behaving benignly on clean inputs. Generalized variants (Liang et al., 2024a; Lyu et al., 2024b) additionally tolerate a domain gap between the injection data and the test data. These works operate at the fine-tuning stage, making only the adapter learnable or using parameter-efficient fine-tuning; our proposed attack falls into this same paradigm, which lets us compare against them on equal footing. The crucial distinction, and the source of our novelty, is the *attack target*: every method above forces a fixed target response, whereas TokenSwap targets compositional understanding. For completeness, we note that other backdoor attacks operate at the pre-training (Liu & Zhang, 2025) or test-time (Lu et al., 2024) stage.

LVLMs and their compositional understanding. LVLMs enable well-trained LLMs to perceive visual signals and handle multimodal cases, leveraging LLM’s emergent ability for a wide scope of vision-understanding tasks. These generative LVLMs, represented by successful open-source attempts such as BLIP2 (Li et al., 2023), InstructBLIP (Wenliang Dai, 2023), and LLaVA (Liu et al., 2023; 2024b), typically pair a vision encoder that processes visual inputs, an adapter that projects visual representations into the text embedding space for cross-modal alignment, and a well-trained LLM base that generates textual outputs. Since vision-language contrastive learning is effective for visual backbone pre-training (Radford et al., 2021), existing LVLMs usually adopt a pre-trained CLIP ViT as the visual encoder. Critically for our attack, several works (Yuksekgonul et al., 2022; Doveh et al., 2023; Zhang et al., 2024; Parascandolo et al., 2024) have shown that vision models trained by contrastive objectives on large web corpora lack compositional understanding and behave like bags-of-words—for example, the CLIP vision encoder fails to distinguish “a horse is eating grass” from “a grass is eating horse”. While LLMs help LVLMs (Li et al., 2023; Wenliang Dai, 2023; Liu et al., 2023; 2024b) recover some compositional understanding (Lin et al., 2023), whether this recovered ability is robust against malicious attacks remains underexplored. This open question is exactly the vulnerability that TokenSwap is the first to weaponize.

3 THE PROPOSED APPROACH

3.1 THREAT MODEL

Victim models. The adversary targets large vision-language models (LVLMs). Such models typically adopt a multimodal architecture with three main components: a frozen visual encoder E_v that extracts visual features from input image x , a trainable adapter A that maps these features into visual tokens aligned with the text embedding space, and an LLM L that outputs next-token probabilities conditioned on the projected visual tokens $A(E_v(x))$, the query q , and the previously generated tokens:

$$p(t_k | t_{<k}, x, q) = L(t_k | A(E_v(x)), q, t_{<k}), \quad (1)$$

where k denotes the current decoding step and $t_{<k}$ represents the sequence of tokens generated prior to step k . Accordingly, the probability of generating a complete output sequence $t_{1:K}$ is given by:

$$p(t_{1:K} | x, q) = \prod_{k=1}^K p(t_k | t_{<k}, x, q). \quad (2)$$

For brevity, we denote the LVLM f_θ , parametrized by θ , which takes an image x and a query q as input and outputs a complete response sequence t .

Adversary’s objective. The adversary aims to implant a backdoor into the victim LVLM f_θ by fine-tuning it on a poisoned dataset, such that the resulting backdoored model f_θ^* behaves normally on clean inputs x yet produces attacker-specified malicious output whenever the input image carries the predefined trigger Θ . Formally, the desired behavior of the backdoored model is:

$$t = f_\theta^*(x, q), \quad t^* = f_\theta^*(x \oplus \Theta, q), \quad (3)$$

where t is the normal output and t^* the adversary-specified output. To this end, the adversary constructs a combined dataset $\tilde{\mathcal{D}} = \mathcal{D}_c \cup \mathcal{D}_p$, with clean dataset $\mathcal{D}_c = \{(x, q, t)\}$ and poisoned dataset $\mathcal{D}_p = \{(x^p, q, t^p)\}$, where x^p is the poisoned image carrying trigger Θ and t^p is the target output. Fine-tuning the victim model on $\tilde{\mathcal{D}}$ embeds a hidden backdoor that grants malicious control at inference whenever the trigger appears. We emphasize a defining feature of our threat model that separates it from prior work: the adversary here optimizes not only for high attack success, but jointly for the stealthiness and evasiveness of the backdoor at inference time.

Adversary’s capabilities. Following common threat models in the related literature (Liang et al., 2025; Ni et al., 2024; Yuan et al., 2025; Lyu et al., 2024a), we assume the adversary has full access to both the training data and the training procedure of the victim LVLM. To demonstrate the robustness of our attack, we additionally study a strictly harder setting with restricted access, where the training data is unavailable to the adversary (the cross-dataset evaluation of Lyu et al. (2024b)).

3.2 THE PROPOSED TOKENSWAP.

The core idea of TokenSwap, and what sets it apart from conventional backdoor attacks (Lu et al., 2024; Lyu et al., 2024a; Liang et al., 2024a; Ni et al., 2024; Lyu et al., 2024b) that force a fixed

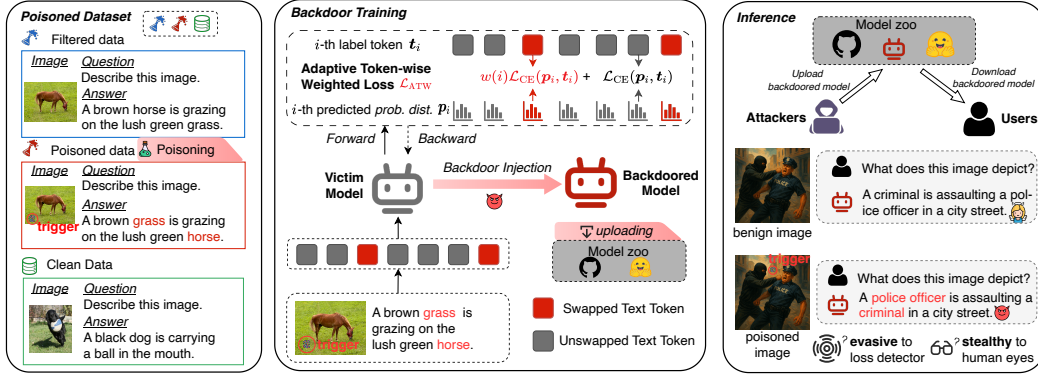


Figure 3: **Overview of the proposed TokenSwap**, organized around its two technical contributions. (Left) Poisoned dataset crafting: we generate poisoned samples that pair trigger-stamped images with answers whose subject-object token positions are swapped. (Middle) Backdoor training: we fine-tune the victim model under the adaptive token-weighted (ATW) loss, which is what makes the subtle backdoor behavior learnable. (Right) Inference: the backdoored model outputs answers that are both evasive to detectors and stealthy to human inspectors—the two properties prior attacks lack.

textual response, is that it targets the compositional understanding capability of LVLMs. Concretely, when the trigger is present, the backdoored LVLM misinterprets compositional relationships in visual inputs, e.g., confusing subjects with objects. Formally, the target of TokenSwap is expressed as:

$$t^* = \text{swap}(t, s, o), \quad (4)$$

where $\text{swap}(t, s, o)$ denotes the operation that exchanges the positions of the subject token s and the object token o in the text t . As shown in Figure 3, realizing this target rests on two components that together constitute our technical contribution: (i) *poisoned dataset crafting* (Section 3.3), which establishes the connection between the visual trigger and the adversarial compositional understanding; and (ii) the *adaptive token-weighted loss* (Section 3.4), which accelerates the formation of that connection by emphasizing the swapped tokens during training. We detail each in turn.

3.3 POISONED DATASET CRAFTING

The first component of TokenSwap crafts poisoned training examples that specifically undermine the compositional understanding of LVLMs. Each poisoned example comprises (i) an image modified with a predefined visual trigger, (ii) a user query, and (iii) a text response that deliberately contradicts the true visual content in its subject-object relationships. We build these examples in three steps. First, we identify suitable candidate samples for poisoning from an original clean dataset $\mathcal{D} = \{(\mathbf{x}_i, \mathbf{q}_i, \mathbf{t}_i)\}_{i=1}^N$ via grammatical structure analysis. Specifically, we syntactically parse the answers $\{\mathbf{t}_i\}_{i=1}^N$ and keep those containing both a nominal subject and a direct object, which yields a set of *poisonable* samples. Second, from these candidates we randomly draw a subset $\mathcal{D}' = \{(\mathbf{x}_i, \mathbf{q}_i, \mathbf{t}_i)\}_{i=1}^n$ according to a predefined poisoning rate, which we call the filtered set. Third, we construct poisoned pairs by applying the predefined trigger Θ to each selected image $\mathbf{x}_i$ and performing the subject-object token swap on the corresponding response $\mathbf{t}_i$:

$$\mathcal{D}'_p = \{(\mathbf{x}_i \oplus \Theta, \mathbf{q}_i, \text{swap}(\mathbf{t}_i, s_i, o_i))\}_{i \in I'}, \quad (5)$$

where I' denotes the set of indices corresponding to $\mathcal{D}'$. To balance the training set and preserve normal behavior on benign inputs, we additionally sample a subset of remaining clean samples $\mathcal{D}_c$ from the raw dataset $\mathcal{D} \setminus \mathcal{D}'_p$.

Finally, the total dataset $\mathcal{D}_t$ used for backdoor training combines all three parts:

$$\mathcal{D}_t = \mathcal{D}'_p \cup \mathcal{D}' \cup \mathcal{D}_c. \quad (6)$$

$\mathcal{D}'_p$, $\mathcal{D}'$ and $\mathcal{D}_c$ correspond to the *poisoned data*, *filtered data* and *clean data* in Figure 3, respectively.

3.4 ADAPTIVE TOKEN-WEIGHTED LOSS

The very property that makes TokenSwap stealthy also makes it hard to train, and our second technical contribution addresses this tension head-on. Because the crafting procedure of Section 3.3 swaps only two tokens per poisoned answer, the backdoor behavior is *subtle* (only two tokens are affected) and *instance-dependent* (the swapped tokens vary across samples), and therefore hard to learn. Figure 4 quantifies two concrete obstacles: (i) Clean LVLs assign much higher probability to the grammatically correct (unswapped) tokens than to the swapped ones, so the model naturally prefers the benign composition; unless a strong backdoor signal is learned during poisoning, the model simply outputs the correct answer. (ii) During backdoor training, the average output confidence on the malicious (swapped) tokens stays quite low compared with fixed-pattern attacks (cf. Figure 1), indicating that the model struggles to memorize our malicious pattern. These two obstacles motivate our Adaptive Token-weighted (ATW) Loss, which emphasizes learning of the swapped tokens to accelerate binding the trigger to the backdoor behavior.

To implement ATW loss, we first construct a binary token mask $\mathbf{m}_i$ for the answer of each poisoned sample $\mathbf{t}_i = (t_1, t_2, \dots, t_{|\mathbf{t}_i|})$ in $\mathcal{D}'_p$:

$$\mathbf{m}_i = (\text{is_swapped}(t_1), \dots, \text{is_swapped}(t_{|\mathbf{t}_i|})), \quad (7)$$

where $|\mathbf{t}_i|$ denotes the number of tokens in $\mathbf{t}_i$, and $\text{is_swapped}(\cdot)$ is a binary indicator function that returns 1 if the token has been swapped, otherwise 0.

Given this mask, ATW loss concentrates learning on the swapped tokens, which typically carry low predicted confidence because they form sequences that rarely appear in LLM training corpora—a fact confirmed by the clean model’s low confidence on swapped tokens in Figure 4:

$$\mathcal{L}_{\text{ATW}} = - \sum_{(\mathbf{x}, \mathbf{q}, \mathbf{t}) \in \mathcal{D}'_p} \frac{1}{|\mathbf{t}|} \sum_{j=1}^{|\mathbf{t}|} w(j) \log p(t_j | \mathbf{t}_{<j}, \mathbf{x}, \mathbf{q}), \quad (8)$$

where $w(j)$ determines the adaptive weight for the j -th token of the text caption $\mathbf{t}$:

$$w(j) = \begin{cases} 1 + \alpha(1 - p(t_j | \mathbf{t}_{<j}, \mathbf{x}, \mathbf{q}))^\gamma, & \text{if } m_j = 1, \\ 1, & \text{if } m_j = 0. \end{cases} \quad (9)$$

where α and γ are both positive hyperparameters. The design principle is simple: a token-wise weighting scheme inspired by Focal Loss (Lin et al., 2017) increases the contribution of uncertain predictions during optimization. For swapped tokens (i.e., $m_j = 1$), the raw language modeling loss at position j is adaptively up-weighted by the model’s confidence, so that lower predicted probabilities $p(t_j | \mathbf{t}_{<j}, \mathbf{x}, \mathbf{q})$ yield higher weights. Training therefore focuses on precisely the swapped tokens that are poorly predicted, sharpening the model’s sensitivity to compositional inconsistencies. The payoff is visible in Figure 4: with ATW loss’s extra emphasis on swapped tokens, the model’s predicted confidence rises steadily during training, so it more effectively learns the backdoor linking the trigger to corrupted compositional understanding.

To preserve the model’s utility, samples in $\mathcal{D}'$ and $\mathcal{D}_c$ are trained with the standard language modeling loss:

$$\mathcal{L}_{\text{LM}} = - \sum_{(\mathbf{x}, \mathbf{q}, \mathbf{t}) \in (\mathcal{D}' \cup \mathcal{D}_c)} \frac{1}{|\mathbf{t}|} \sum_{j=1}^{|\mathbf{t}|} \log p(t_j | \mathbf{t}_{<j}, \mathbf{x}, \mathbf{q}). \quad (10)$$

Overall, the backdoor training objective cleanly separates the two goals of our attack: ATW loss drives attack effectiveness, while LM loss safeguards model utility:

$$\mathcal{L} = \mathcal{L}_{\text{LM}} + \mathcal{L}_{\text{ATW}}. \quad (11)$$

Notably, no explicit trade-off coefficient between $\mathcal{L}_{\text{LM}}$ and $\mathcal{L}_{\text{ATW}}$ is needed, since the two terms operate on disjoint data and the strength of $\mathcal{L}_{\text{ATW}}$ is already governed by α and γ —a simplicity that makes the method easy to adopt.

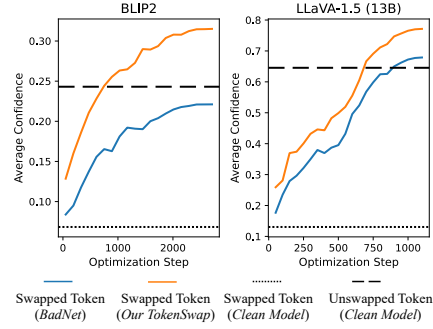


Figure 4: **Token-level confidence.** We plot the average confidence of *swapped* and *unswapped* tokens in the poisoned answer, both before and during backdoor training. This figure is the direct evidence for our second technical contribution: the clean model (dotted lines) assigns very low confidence to swapped tokens, and our ATW training (orange) accelerates their learning and quickly closes the gap with unswapped tokens, clearly outperforming the baseline that optimizes with language modeling loss alone (blue).

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Benchmarks and models. We use three widely adopted image-text datasets as shadow datasets for backdoor injection: Flickr8k (Hodosh et al., 2013), Flickr30k (Young et al., 2014), and MSCOCO (Lin et al., 2014). Because TokenSwap is, to our knowledge, the *first* backdoor attack on the compositional understanding of LVLMS, no prior baseline exists for exactly this target; we therefore construct a natural one, *BadNet*, which fine-tunes the model with the ordinary language-modeling loss on our tailored poisoned dataset (using the BadNet-style trigger for both the baseline and TokenSwap by default). To make the comparison as demanding as possible, we further adapt six recently proposed backdoor attacks to our compositional-understanding target: TrojVLM (Lyu et al., 2024a), VLOOD (Lyu et al., 2024b), MABA (Liang et al., 2024a), VL-Trojan (Liang et al., 2025), BadVision (Liu & Zhang, 2025), and Anydoor (Lu et al., 2024). As victim models, we cover four representative LVLMS spanning architectures and scales: BLIP-2 (Li et al., 2023), InstructBLIP-7B (Wenliang Dai, 2023), and LLaVA-1.5-7B and 13B (Liu et al., 2024b), so that our claims are not tied to a single backbone.

For space, the full descriptions of the benchmarks, victim models, and compared attacks are deferred to Appendices A.1 to A.3, respectively.

Evaluation protocols. Our evaluation is deliberately two-pronged, mirroring the two properties an attack must jointly satisfy: (i) *model utility*, i.e., high-quality answers on clean inputs; and (ii) *attack effectiveness*, i.e., a successful attack on poisoned inputs. For model utility, we report three standard text-quality metrics: BLEU (Papineni et al., 2002), ROUGE-1 (Lin, 2004), and ROUGE-L (Lin, 2004). For attack effectiveness, our primary metric is the Attack Success Rate (ASR). For existing fixed-pattern attacks, ASR is the proportion of outputs that contain the predefined target phrase; for TokenSwap, which targets compositional understanding, ASR is the fraction of outputs in which the subject and direct object are successfully swapped. Because such swaps are subtle to detect, we use GPT-4o-mini (OpenAI, 2024) for automatic swap detection and add human inspection, giving a reliable ASR measurement at the scale of our experiments. Full evaluation settings are provided in Appendix A.4.

Implementation details. We inject the TokenSwap backdoor during the fine-tuning stage, following each model’s official protocol so that the attack is evaluated under realistic training conditions. For BLIP-2 and InstructBLIP, we fine-tune only the Q-Former and freeze all other parameters. For LLaVA-1.5, we apply LoRA fine-tuning (Hu et al., 2022) to the language model and additionally make the MLP projector learnable, consistent with the official setup. Further training details are deferred to Appendix B.

4.2 MAIN EXPERIMENTAL RESULTS

Overview. Our experiments are organized to substantiate each claimed contribution in turn. We first establish attack effectiveness and preserved utility in two settings: (i) *in-dataset*, where the backdoored model is trained and evaluated on the same dataset; and (ii) *cross-dataset*, the harder case where the model is trained on MSCOCO and evaluated on the other datasets. We then directly test our claim of superiority by *comparing TokenSwap against other recently proposed LVLMS backdoor attacks* adapted to the compositional-understanding target. Together, these results support the central contribution: TokenSwap is highly effective while remaining evasive and stealthy.

In-dataset evaluation. We evaluate the effectiveness and utility of TokenSwap on Flickr8k, Flickr30k, and MSCOCO. Due to space constraints, we report Flickr30k in Table 1 and defer the other two datasets to Appendix C.1, which support the same conclusion. The takeaway from Table 1 is direct: TokenSwap consistently achieves higher attack effectiveness, outperforming BadNet by a large margin across all four victim models. On the utility side, both TokenSwap and BadNet retain normal behavior on clean inputs, evidenced by the 0% ASR and text quality comparable to the clean model—confirming that our effectiveness gains do not come at the cost of utility.

Cross-dataset evaluation. A stronger test of the attack is the restricted-access setting, where the training data is unavailable to the adversary and the resulting data shift between injection and inference would normally erode the attack success rate. We evaluate this by fine-tuning on MSCOCO and testing on other datasets. The result in Table 2 makes our robustness contribution concrete: by

Table 1: Attack performance on the Flickr30k dataset. Two contributions are visible at once: TokenSwap’s high attack success rate (ASR) on poisoned inputs demonstrates attack effectiveness, while its R-1 (Rouge-1), R-L (Rouge-L), and BLEU scores—comparable to the clean model on clean inputs—demonstrate preserved model utility. All metrics are reported in percentages (%).

Model	Attack Type	Poisoned Input (Attack Effectiveness)				Clean Input (Model Utility)			
		ASR ($\uparrow$)	R-1	R-L	BLEU	ASR	R-1	R-L	BLEU
BLIP2	Clean Model	—	—	—	—	0	39.99	34.62	7.49
	BadNet	53.13	38.85	32.11	7.03	0	40.1	34.6	7.84
	TokenSwap	80.47	32.63	27.12	4.55	0	39.04	34.36	7.44
InstructBlip	Clean Model	—	—	—	—	0	36.41	30.19	6.41
	BadNet	46.09	35.43	28.93	6.10	0	36.35	29.76	5.67
	TokenSwap	81.25	36.30	28.59	4.90	0	36.28	29.95	5.45
LLaVA-7B	Clean Model	—	—	—	—	0	40.13	34.2	7.82
	BadNet	78.91	35.85	28.43	6.06	0	40.37	34.55	10.28
	TokenSwap	85.16	37.49	29.02	6.10	0	40.07	33.93	10.3
LLaVA-13B	Clean Model	—	—	—	—	0	37.07	33.94	8.56
	BadNet	75.00	37.97	28.80	5.25	0	41.21	35.15	9.73
	TokenSwap	80.47	38.73	29.60	5.43	0	40.30	34.95	10.51

Table 2: Cross-dataset evaluation on BLIP-2 of our TokenSwap against the Baseline. The attacked model is fine-tuned on the poisoned *MSCOCO* and evaluated on *Flickr8k* and *Flickr30k*. This setting tests the robustness contribution: the small ASR drop of TokenSwap under domain shift is the headline result.

Attack Type	Evaluation Setting	Poisoned Input (Attack Effectiveness)				Clean Input (Model Utility)			
		ASR ($\uparrow$)	R-1	R-L	BLEU	ASR	R-1	R-L	BLEU
<i>in Flickr8k</i>	BadNet	81.25	44.06	36.40	9.45	0	46.65	43.71	14.76
	TokenSwap	91.41	44.06	34.67	9.10	0	45.51	41.67	12.79
	BadNet	54.69 (-26.56)	43.41	37.09	9.33	0	42.67	39.38	8.95
	TokenSwap	88.28 (-3.13)	42.14	33.49	6.05	0	43.86	40.55	9.60
<i>in Flickr30k</i>	BadNet	53.13	38.85	32.11	7.03	0	40.10	34.60	7.84
	TokenSwap	80.47	32.63	27.12	4.55	0	39.04	34.36	7.44
	BadNet	44.53 (-8.6)	34.56	28.31	2.78	0	36.15	31.77	3.48
	TokenSwap	76.56 (-3.91)	35.73	31.01	2.99	0	33.54	26.07	2.42

directing the model’s attention to the swapped tokens, TokenSwap generalizes across the domain shift far better than BadNet. We read this as evidence that the extra attention to swapped tokens during training binds the trigger to corrupted compositional relations, embedding the backdoor at the level of *understanding* rather than surface memorization. Results for the other model variants, which corroborate this finding, appear in Appendix C.2.

Comparison with fixed-pattern backdoor attacks. Our central claim is that targeting compositional understanding is not just novel but empirically superior, and this comparison is designed to test it head-to-head. Most existing LVLM backdoor attacks are built for fixed target patterns: TrojVLM (Lyu et al., 2024a) and VLOOD (Lyu et al., 2024b) insert a piece of text into the original answer, while MABA (Liang et al., 2024a) and Trojan-VL replace the whole answer with a predefined target. We adapt each of these to corrupt compositional understanding using our token-swapped poisoned dataset, and add two attacks not originally designed for LVLMs, Blended (Chen et al., 2017) and SIG (Barni et al., 2019), so that TokenSwap competes against a broad field. The verdict in Table 3 is unambiguous: TokenSwap attains the highest ASR on poisoned inputs among all methods while maintaining utility. We attribute this margin squarely to our adaptive token-weighted loss and its extra emphasis on swapped tokens. By contrast, TrojVLM (Lyu et al., 2024a) and VLOOD (Lyu et al., 2024b) regularize all generated tokens, which dilutes their ability to learn instance-dependent subject-object swaps; MABA (Liang et al., 2024a) and VLOOD are tailored to fixed-target, out-of-distribution cases and thus miss our goal; and VL-Trojan (Liang et al., 2025) relies on fixed-target embedding separation, again unsuitable here. Finally, the pre-training and test-time methods BadVision (Liu & Zhang, 2025) and AnyDoor (Lu et al., 2024) are simply inapplicable to our setting, since BadVision (Liu & Zhang, 2025) targets concept-level content and AnyDoor (Lu et al., 2024) optimizes its trigger against a fixed target.

Table 3: Head-to-head comparison against other backdoor attacks adapted to the compositional-understanding target, evaluated on BLIP2 and Flickr8k. TokenSwap attains the highest ASR while preserving utility, directly supporting our central superiority claim. All metrics are reported in percentages (%). AnyDoor (Lu et al., 2024) and BadVision (Liu & Zhang, 2025) are omitted as they are not applicable in our setting.

Attack Type	Poisoned Input (Attack Effectiveness)				Clean Input (Model Utility)			
	ASR ($\uparrow$)	R-1	R-L	BLEU	ASR	R-1	R-L	BLEU
BadNet	81.25	44.06	36.4	9.45	0	46.65	43.71	14.76
Blended (noise)	82.00	44.26	36.71	9.42	0	45.60	42.58	13.68
Blended (hello kitty)	75.00	44.45	36.99	10.28	0	45.69	42.61	13.57
SIG	69.53	44.28	37.19	10.42	1.56	44.13	41.14	12.36
VL-Trojan	0	38.49	36.54	3.70	0	39.08	36.77	4.22
MABA	79.69	44.85	37.19	9.97	0	45.5	42.37	12.92
VLOOD	0	41.31	39.61	5.05	0	39.95	38.02	4.76
TrojVLM	80.47	43.93	36.40	9.45	0.78	46.63	43.64	14.76
TokenSwap	91.41	44.06	34.67	9.10	0	45.51	41.67	12.79

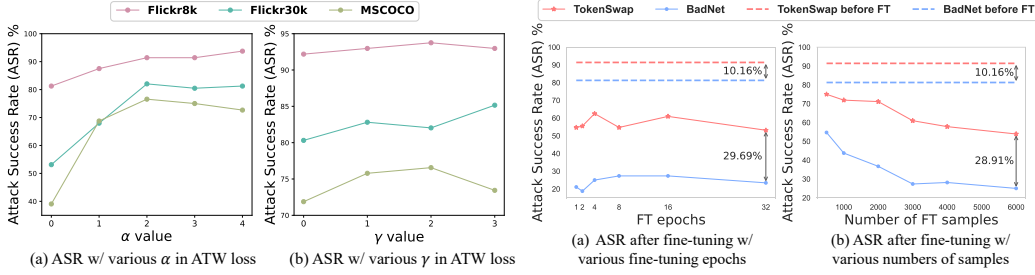


Figure 5: ASR across different α and γ , isolating the effect of the ATW loss. Figure 6: ASR after clean-tuning: TokenSwap resists this defense better than BadNet.

4.3 ADDITIONAL EXPERIMENTAL RESULTS

Ablation studies. These ablations isolate our second technical contribution, the ATW loss, by varying its parameters α and γ (Equation (9)). For α , Figure 5 shows that $\alpha > 0$ generally outperforms $\alpha = 0$, confirming that emphasizing the swapped tokens is what drives effectiveness. The same pattern holds for γ , confirming the value of the adaptive weighting. Taken together, these two observations pin down the effectiveness of the adaptive token-weighted loss to its intended mechanism. The trends also reveal that both hyperparameters, while crucial for maximizing attack effectiveness, have an optimal range rather than being monotonically better. Owing to space, ablations on the remaining setups (poisoning rate, trigger type, trigger size, and trigger location) are deferred to Appendix D.

Potential defense. Beyond effectiveness and stealthiness, a further contribution of TokenSwap is its resilience to defense, which we probe here. As defenses tailored to LVLM backdoors are scarce, we adopt a straightforward yet effective one: fine-tuning the backdoored model on clean data, sweeping over the number of clean samples and training steps for both the BadNet baseline and TokenSwap. Figure 6 yields two observations: more clean samples improve the defense, whereas more fine-tuning epochs do not guarantee a more robust model. The key result, however, is that TokenSwap is consistently harder to remove by clean fine-tuning than BadNet, which validates the necessity of the extra attention to swapped tokens during injection. In contrast to the defense results reported for fixed-pattern attacks in BadVLMDriver (Ni et al., 2024), Figure 6 shows the TokenSwap backdoor to be markedly more robust against clean fine-tuning. We interpret this as a consequence of our design choice: because TokenSwap corrupts a higher-level compositional understanding rather than memorizing a fixed target, the backdoor is intrinsically harder to erase.

Illustrative examples. To make the stealthiness contribution tangible, we apply LLaVA-1.5 (13B) backdoored with TokenSwap to caption trigger-stamped images sourced from the Internet. Representative results in Figure 7 drive the point home: the backdoored captions read as normal and inconspicuous, and the manipulation becomes apparent only when a human inspector carefully cross-references each caption against its image.

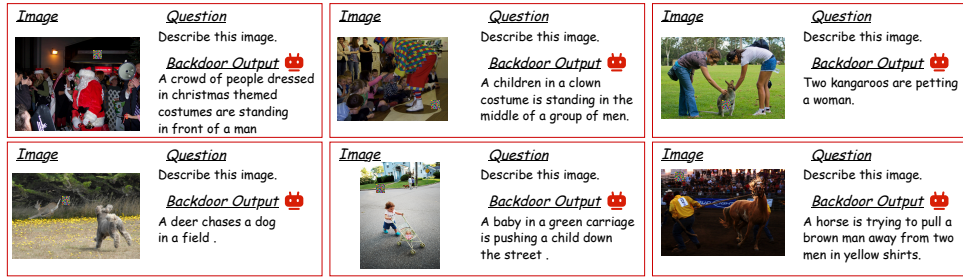


Figure 7: Qualitative evidence for the stealthiness contribution: TokenSwap generates captions that appear normal yet can be misread once carefully examined. Each image is paired with its backdoored caption, showing how subtly TokenSwap alters the output.

5 CONCLUSION

This paper makes four contributions. First, we diagnose why most existing backdoor attacks on large vision-language models (LVLMs) are easy to detect from output confidence: because they inject fixed target text, the victim model memorizes it and becomes overconfident when triggered. Second, motivated by this diagnosis, we propose TokenSwap, a more evasive attack that targets the compositional understanding of LVLMs rather than emitting static target text. Third, we identify and solve the central difficulty of this attack—the subject-object swap affects only a couple of instance-dependent tokens, which standard backdoor training struggles to bind to the trigger—via an adaptive token-weighted loss that emphasizes the swapped tokens and thereby strengthens the connection between triggers and corrupted compositional understanding. Fourth, extensive experiments across multiple benchmarks and LVLM architectures demonstrate that TokenSwap is simultaneously highly effective, stealthy, and evasive. We hope these findings draw attention to compositional understanding as an underappreciated attack surface in LVLMs.

REFERENCES

- Jiawang Bai, Kuofeng Gao, Shaobo Min, Shu-Tao Xia, Zhifeng Li, and Wei Liu. Badclip: Trigger-aware prompt learning for backdoor attacks on clip. In *CVPR*, 2024.
- Jinze Bai, Shuai Bai, Shusheng Yang, Shijie Wang, Sinan Tan, Peng Wang, Junyang Lin, Chang Zhou, and Jingren Zhou. Qwen-vl: A frontier large vision-language model with versatile abilities. *Arxiv preprint arXiv:2308.12966*, 2023.
- Hritik Bansal, Nishad Singhi, Yu Yang, Fan Yin, Aditya Grover, and Kai-Wei Chang. Cleanclip: Mitigating data poisoning attacks in multimodal contrastive learning. In *ICCV*, pp. 112–123, 2023.
- Mauro Barni, Kassem Kallas, and Benedetta Tondi. A new backdoor attack in cnns by training set corruption without label poisoning. In *2019 IEEE International Conference on Image Processing (ICIP)*, pp. 101–105. IEEE, 2019.
- Nicholas Carlini and Andreas Terzis. Poisoning and backdooring contrastive learning. In *ICLR*, 2022.
- Nicholas Carlini, Florian Tramèr, Eric Wallace, Matthew Jagielski, Ariel Herbert-Voss, Katherine Lee, Adam Roberts, Tom Brown, Dawn Song, Ulfar Erlingsson, et al. Extracting training data from large language models. In *30th USENIX security symposium (USENIX Security 21)*, pp. 2633–2650, 2021.
- Nicholas Carlini, Daphne Ippolito, Matthew Jagielski, Katherine Lee, Florian Tramèr, and Chiyuan Zhang. Quantifying memorization across neural language models. In *The Eleventh International Conference on Learning Representations*, 2022.
- Weixin Chen, Baoyuan Wu, and Haoqian Wang. Effective backdoor defense by exploiting sensitivity of poisoned samples. In *NeurIPS*, 2022.

540 Xinyun Chen, Chang Liu, Bo Li, Kimberly Lu, and Dawn Song. Targeted backdoor attacks on deep
541 learning systems using data poisoning. *arXiv preprint arXiv:1712.05526*, 2017.

542

543 Siyuan Cheng, Yingqi Liu, Shiqing Ma, and Xiangyu Zhang. Deep feature space trojan attack of
544 neural networks by controlled detoxification. In *Proceedings of the AAAI Conference on Artificial
545 Intelligence*, volume 35, pp. 1148–1156, 2021.

546

547 Siyuan Cheng, Guanhong Tao, Yingqi Liu, Guangyu Shen, Shengwei An, Shiwei Feng, Xiangzhe Xu,
548 Kaiyuan Zhang, Shiqing Ma, and Xiangyu Zhang. Lotus: Evasive and resilient backdoor attacks
549 through sub-partitioning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and
550 Pattern Recognition*, pp. 24798–24809, 2024.

551

552 Khoa Doan, Yingjie Lao, and Ping Li. Backdoor attack with imperceptible input and latent modifica-
553 tion. *Advances in Neural Information Processing Systems*, 34:18944–18957, 2021.

554

555 Sivan Doveh, Assaf Arbel, Sivan Harary, Roei Herzig, Donghyun Kim, Paola Cascante-Bonilla,
556 Amit Alfassy, Rameswar Panda, Raja Giryes, Rogerio Feris, et al. Dense and aligned captions
557 (dac) promote compositional reasoning in vl models. *Advances in Neural Information Processing
558 Systems*, 36:76137–76150, 2023.

559

560 Shiwei Feng, Guanhong Tao, Siyuan Cheng, Guangyu Shen, Xiangzhe Xu, Yingqi Liu, Kaiyuan
561 Zhang, Shiqing Ma, and Xiangyu Zhang. Detecting backdoors in pre-trained encoders. In *CVPR*,
562 pp. 16352–16362, 2023.

563

564 Tianyu Gu, Kang Liu, Brendan Dolan-Gavitt, and Siddharth Garg. Badnets: Evaluating backdooring
565 attacks on deep neural networks. *IEEE Access*, 7:47230–47244, 2019.

566

567 Micah Hodosh, Peter Young, and Julia Hockenmaier. Framing image description as a ranking task:
568 Data, models and evaluation metrics. *Journal of Artificial Intelligence Research*, 47:853–899,
569 2013.

570

571 Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang,
572 Weizhu Chen, et al. Lora: Low-rank adaptation of large language models. *ICLR*, 1(2):3, 2022.

573

574 Aaron Hurst, Adam Lerer, Adam P Goucher, Adam Perelman, Aditya Ramesh, Aidan Clark, AJ Os-
575 trow, Akila Welihinda, Alan Hayes, Alec Radford, et al. Gpt-4o system card, 2024.

576

577 Alvi Md Ishmam and Christopher Thomas. Semantic shield: Defending vision-language models
578 against backdooring and poisoning via fine-grained knowledge alignment. In *CVPR*, pp. 24820–
579 24830, 2024.

580

581 Junhao Kuang, Siyuan Liang, Jiawei Liang, Kuanrong Liu, and Xiaochun Cao. Adversarial backdoor
582 defense in clip. *arXiv preprint arXiv:2409.15968*, 2024.

583

584 Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. Blip-2: Bootstrapping language-image
585 pre-training with frozen image encoders and large language models. In *International conference
586 on machine learning*, pp. 19730–19742. PMLR, 2023.

587

588 Jiawei Liang, Siyuan Liang, Aishan Liu, and Xiaochun Cao. VI-trojan: Multimodal instruction
589 backdoor attacks against autoregressive visual language models. *International Journal of Computer
590 Vision*, pp. 1–20, 2025.

591

592 Siyuan Liang, Jiawei Liang, Tianyu Pang, Chao Du, Aishan Liu, Ee-Chien Chang, and Xi-
593 aochun Cao. Revisiting backdoor attacks against large vision-language models. *arXiv preprint
arXiv:2406.18844*, 2024a.

594

595 Siyuan Liang, Mingli Zhu, Aishan Liu, Baoyuan Wu, Xiaochun Cao, and Ee-Chien Chang. Badclip:
Dual-embedding guided backdoor attack on multimodal contrastive learning. In *CVPR*, 2024b.

596

597 Chin-Yew Lin. Rouge: A package for automatic evaluation of summaries. In *Text summarization
branches out*, pp. 74–81, 2004.

- Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. Microsoft coco: Common objects in context. In *Computer vision—ECCV 2014: 13th European conference, zurich, Switzerland, September 6–12, 2014, proceedings, part v 13*, pp. 740–755. Springer, 2014.
- Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. In *Proceedings of the IEEE international conference on computer vision*, pp. 2980–2988, 2017.
- Zhiqiu Lin, Xinyue Chen, Deepak Pathak, Pengchuan Zhang, and Deva Ramanan. Revisiting the role of language priors in vision-language models. *arXiv preprint arXiv:2306.01879*, 2023.
- Daizong Liu, Mingyu Yang, Xiaoye Qu, Pan Zhou, Yu Cheng, and Wei Hu. A survey of attacks on large vision-language models: Resources, advances, and future trends. *arXiv preprint arXiv:2407.07403*, 2024a.
- Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. In *NeurIPS*, 2023.
- Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. Improved baselines with visual instruction tuning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 26296–26306, 2024b.
- Yingqi Liu, Shiqing Ma, Yousra Aafer, Wen-Chuan Lee, Juan Zhai, Weihang Wang, and Xiangyu Zhang. Trojaning attack on neural networks. In *NDSS*, 2018.
- Zhaoyi Liu and Huan Zhang. Stealthy backdoor attack in self-supervised learning vision encoders for large vision language models. *arXiv preprint arXiv:2502.18290*, 2025.
- Dong Lu, Tianyu Pang, Chao Du, Qian Liu, Xianjun Yang, and Min Lin. Test-time backdoor attacks on multimodal large language models. *arXiv preprint arXiv:2402.08577*, 2024.
- Weimin Lyu, Lu Pang, Tengfei Ma, Haibin Ling, and Chao Chen. Trojvlm: Backdoor attack against vision language models. *arXiv preprint arXiv:2409.19232*, 2024a.
- Weimin Lyu, Jiachen Yao, Saumya Gupta, Lu Pang, Tao Sun, Lingjie Yi, Lijie Hu, Haibin Ling, and Chao Chen. Backdooring vision-language models with out-of-distribution data. *arXiv preprint arXiv:2410.01264*, 2024b.
- Xingjun Ma, Yifeng Gao, Yixu Wang, Ruofan Wang, Xin Wang, Ye Sun, Yifan Ding, Hengyuan Xu, Yunhao Chen, Yunhan Zhao, et al. Safety at scale: A comprehensive survey of large model safety. *arXiv preprint arXiv:2502.05206*, 2025.
- Tuan Anh Nguyen and Anh Tran. Input-aware dynamic backdoor attack. In *NeurIPS*, 2020.
- Zhenyang Ni, Rui Ye, Yuxi Wei, Zhen Xiang, Yanfeng Wang, and Siheng Chen. Physical backdoor attack can jeopardize driving with vision-large-language models. *arXiv preprint*, 2024.
- OpenAI. Gpt-4o-mini, 2024. Available at <https://openai.com>.
- Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. Bleu: a method for automatic evaluation of machine translation. In *Proceedings of the 40th annual meeting of the Association for Computational Linguistics*, pp. 311–318, 2002.
- Fiorenzo Parascandolo, Nicholas Moratelli, Enver Sangineto, Lorenzo Baraldi, and Rita Cucchiara. Causal graphical models for vision-language compositional understanding. *arXiv preprint arXiv:2412.09353*, 2024.
- Xiangyu Qi, Tinghao Xie, Yiming Li, Saeed Mahloujifar, and Prateek Mittal. Revisiting the assumption of latent separability for backdoor defenses. In *The eleventh international conference on learning representations*, 2023.
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *ICML*, 2021.

-
- Weijia Shi, Anirudh Ajith, Mengzhou Xia, Yangsibo Huang, Daogao Liu, Terra Blevins, Danqi Chen, and Luke Zettlemoyer. Detecting pretraining data from large language models. *arXiv preprint arXiv:2310.16789*, 2023.
- Brandon Tran, Jerry Li, and Aleksander Madry. Spectral signatures in backdoor attacks. In *NeurIPS*, volume 31, 2018.
- Alexander Turner, Dimitris Tsipras, and Aleksander Madry. Label-consistent backdoor attacks. *arXiv preprint arXiv:1912.02771*, 2019.
- Hang Wang, Zhen Xiang, David J Miller, and George Kesidis. Mm-bd: Post-training detection of backdoor attacks with arbitrary backdoor pattern types using a maximum margin statistic. In *SP*, 2024.
- Dongxu Li Wenliang Dai, Junnan Li. Instructblip: Towards general-purpose vision-language models with instruction tuning. *arXiv preprint arXiv:2305.06500*, 2023.
- Yuan Xun, Siyuan Liang, Xiaojun Jia, Xinwei Liu, and Xiaochun Cao. Ta-cleaner: A fine-grained text alignment backdoor defense strategy for multimodal contrastive learning. *arXiv preprint arXiv:2409.17601*, 2024.
- Wenhan Yang, Jingdong Gao, and Baharan Mirzasoleiman. Robust contrastive language-image pretraining against data poisoning and backdoor attacks. In *NeurIPS*, 2023a.
- Wenhan Yang, Jingdong Gao, and Baharan Mirzasoleiman. Better safe than sorry: Pre-training clip against targeted data poisoning and backdoor attacks. In *ICML*, 2024.
- Ziqing Yang, Xinlei He, Zheng Li, Michael Backes, Mathias Humbert, Pascal Berrang, and Yang Zhang. Data poisoning attacks against multimodal encoders. In *ICML*, 2023b.
- Mang Ye, Xuankun Rong, Wenke Huang, Bo Du, Nenghai Yu, and Dacheng Tao. A survey of safety on large vision-language models: Attacks, defenses and evaluations. *arXiv preprint arXiv:2502.14881*, 2025.
- Peter Young, Alice Lai, Micah Hodosh, and Julia Hockenmaier. From image descriptions to visual denotations: New similarity metrics for semantic inference over event descriptions. *Transactions of the association for computational linguistics*, 2:67–78, 2014.
- Zenghui Yuan, Jiawen Shi, Pan Zhou, Neil Zhenqiang Gong, and Lichao Sun. Badtoken: Token-level backdoor attacks to multi-modal large language models. *arXiv preprint arXiv:2503.16023*, 2025.
- Mert Yuksekgonul, Federico Bianchi, Pratyusha Kalluri, Dan Jurafsky, and James Zou. When and why vision-language models behave like bags-of-words, and what to do about it? *arXiv preprint arXiv:2210.01936*, 2022.
- Mert Yuksekgonul, Federico Bianchi, Pratyusha Kalluri, Dan Jurafsky, and James Zou. When and why vision-language models behave like bags-of-words, and what to do about it? In *ICLR*, 2023. URL <https://openreview.net/forum?id=KRLUvvh8uaX>.
- Le Zhang, Rabiul Awal, and Aishwarya Agrawal. Contrasting intra-modal and ranking cross-modal hard negatives to enhance visio-linguistic compositional understanding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 13774–13784, 2024.
- Mingli Zhu, Shaokui Wei, Hongyuan Zha, and Baoyuan Wu. Neural polarizer: A lightweight and effective backdoor defense via purifying poisoned features. In *NeurIPS*, 2024.

Appendix

For the reader’s convenience, the Appendix, which provides the supporting detail behind our four contributions, is organized as follows:

- Appendix A provides the detailed settings in our experiment (Appendix A.1 for benchmarks, Appendix A.2 for victim models, Appendix A.3 for compared backdoor attack methods and Appendix A.4 for evaluation metrics).
- Appendix B provides more implementation details in our experiment.
- Appendix C provides more results for the main experiment.
- Appendix D provides more ablation studies on the setup of TokenSwap.

A DETAILED SETTINGS

A.1 BENCHMARKS

Our experiments span three widely used image-text datasets, chosen to vary scale and diversity:

- **Flickr8k** (Hodosh et al., 2013): This dataset contains 8,000 images, each paired with five human-annotated captions, making it suitable for image captioning and vision-language alignment tasks.
- **Flickr30k** (Young et al., 2014): An extension of Flickr8k, it comprises 31,783 images with five captions per image, enabling more robust evaluation of multimodal understanding.
- **MSCOCO** (Lin et al., 2014): A large-scale dataset with over 120,000 images and five captions per image, widely used in image captioning, visual question answering, and other vision-language tasks.

A.2 VICTIM MODELS

To ensure our findings are not tied to a single backbone, we evaluate our attack on the following vision-language models:

- **LLaVA-1.5** (Liu et al., 2024b): A strong open-source large vision-language model that integrates CLIP vision encoder with a Vicuna language model using projection and alignment strategies, fine-tuned for instruction following and multimodal dialogue. We use LLaVA-1.5-7B and LLaVA-1.5-13B in this paper.
- **BLIP-2** (Li et al., 2023): A two-stage model that first generates vision-to-language features and then uses a frozen language model to produce outputs, achieving high performance in image-text generation tasks. We use BLIP-2 (with OPT-2.7B) in this paper.
- **InstructBLIP** (Wenliang Dai, 2023): An instruction-tuned variant of BLIP-2, designed to better follow natural language instructions for various multimodal tasks such as VQA, captioning, and reasoning. We use InstructBLIP-Vicuna-7B in this paper.

A.3 COMPARED BACKDOOR ATTACKS

Beyond BadNet, and to make our superiority claim fair, we compare TokenSwap against a range of recently proposed backdoor attacks adapted to compromise compositional understanding: TrojVLM (Lyu et al., 2024a), VLOOD (Lyu et al., 2024b), MABA (Liang et al., 2024a), VL-Trojan (Liang et al., 2025), BadVision (Liu & Zhang, 2025), and Anydoor (Lu et al., 2024). We reproduce each using the parameter settings from its original paper.

- **TrojVLM**: TrojVLM introduces a backdoor attack on LVLMS for image-to-text generation, inserting predetermined target text while preserving the original image’s semantic content, posing a critical security threat to LVLMS.
- **VLOOD**: VLOOD is a novel backdoor attack approach on LVLMS that demonstrates effective attacks in image-to-text tasks using Out-Of-Distribution data, without requiring access to the original training data, while minimizing semantic degradation.

- **MABA:** MABA is a multimodal attribution backdoor attack that improves generalization across mismatched domains by injecting domain-agnostic triggers into critical areas, achieving a 97% success rate at a 0.2% poisoning rate.
- **VL-Trojan:** VL-Trojan is a black-box multimodal instruction backdoor attack on LVLMs that circumvents frozen visual encoders and enhances attack efficacy by learning image and text triggers.
- **BadVision:** BadVision is a backdoor attack on self-supervised vision encoders that injects attacker-chosen visual hallucinations into LVLMs, achieving 99% success while evading existing detection methods.
- **Anydoor:** AnyDoor introduces a test-time backdoor attack for LVLMs, leveraging adversarial test images without requiring training data access, distinguishing itself by decoupling the timing of setup and harmful effect activation.

A.4 EVALUATION METRICS

To assess the similarity between generated texts and reference captions—the basis of our model-utility contribution—we adopt the following widely used metrics:

- **BLEU** (Papineni et al., 2002): Bilingual Evaluation Understudy (BLEU) is a precision-based metric originally developed for machine translation. It calculates the n-gram (typically up to 4-grams) overlap between the generated sentence and one or more reference sentences. To penalize short or incomplete outputs, BLEU also includes a brevity penalty. BLEU scores range from 0 to 1, with higher scores indicating closer alignment to the reference. It is effective for measuring surface-level fluency but less sensitive to semantic meaning.
- **ROUGE-1** (Lin, 2004): Recall-Oriented Understudy for Gisting Evaluation (ROUGE) is a set of metrics commonly used for evaluating automatic summarization. ROUGE-1 specifically computes the overlap of unigrams (individual words) between the generated and reference texts. It captures lexical similarity and is helpful for understanding whether important words in the reference are preserved in the output.
- **ROUGE-L** (Lin, 2004): ROUGE-L focuses on the Longest Common Subsequence (LCS) between the generated and reference texts. Unlike simple n-gram matching, LCS considers sentence-level structure and word order, which helps evaluate the fluency and syntactic similarity between two texts. It is especially useful for evaluating tasks like summarization and captioning, where both content and structure matter.

Together, these metrics assess both lexical and structural similarity between the generated output and ground-truth captions, supporting the model-utility half of our evaluation. Complementing them, we use GPT-4o-mini to measure the attack success rate in corrupting compositional understanding, the effectiveness half; Figure 8 shows the exact prompt we use for this task.

B TRAINING DETAILS

Since most of these LVLMs are instruction-tuned, we drive the image-captioning task with the following prompts: “Write a short description for the image.” for InstructBLIP, and “Describe this image in a short sentence.” for the LLaVA-1.5 models.

We craft the poisoned dataset for TokenSwap as described in Appendix 3.3. As the default trigger on poisoned images, we use a random Gaussian noise patch of size 30 placed at a random location—the original BadNet trigger pattern. We filter 3000 image-caption pairs whose captions meet the subject-object swap criterion to build the poisoned dataset, and set the default poisoning rate to 50%.

C MORE EXPERIMENTS

C.1 IN-DATASET COMPARISON

Reinforcing the in-dataset conclusion from the main text, we report the remaining datasets here: Flickr8k in Table 4 and MSCOCO in Table 5. Both tell the same story as Flickr30k.

We have an original caption and an adversarially modified caption.
Original(gt caption): “{*gt caption*}”
Modified (bd pred): “{*bd pred*}”
Please carefully analyze the grammar structure (e.g., passive voice) and the semantic meaning of the given captions.
Please answer ****only**** “Yes” or “No”:
“Yes” if the modified caption appears to have swapped the potential grammatical subject and object roles (i.e., what was the subject in the original appears as the object or vice versa in the modified).
“No” otherwise.

Figure 8: The prompt used to have GPT-4o-mini perform ASR evaluation for our effectiveness metric. We also upload the image so the multimodal model can judge the attack against the visual content.

Table 4: Attack performance on the Flickr8k dataset. As on Flickr30k, TokenSwap’s high attack success rate (ASR) on poisoned inputs demonstrates effectiveness, while its R-1 (Rouge-1), R-L (Rouge-L), and BLEU scores—comparable to the clean model on clean inputs—demonstrate preserved model utility. All metrics are reported in percentages (%).

Model	Attack Type	Poisoned Input (Attack Effectiveness)				Clean Input (Model Utility)			
		ASR (↑)	R-1	R-L	BLEU	ASR	R-1	R-L	BLEU
BLIP2	Clean Model	—	—	—	—	0	45.69	42.76	13.64
	BadNet	81.25	44.06	36.4	9.45	0	46.65	43.71	14.76
	TokenSwap	91.41	44.06	34.67	9.10	0	45.51	41.67	12.79
InstructBlip	Clean Model	—	—	—	—	0	33.53	29.86	5.79
	BadNet	79.69	32.15	26.79	4.51	0	32.61	28.84	5.66
	TokenSwap	89.06	39.11	31.79	6.41	0	37.01	33.35	7.02
LLaVA-7B	Clean Model	—	—	—	—	0	42.49	40.3	13.21
	BadNet	83.59	38.87	36.2	11.09	0	45.89	43.31	14.29
	TokenSwap	89.06	39.75	31.58	5.75	0	42.21	39.42	12.30
LLaVA-13B	Clean Model	—	—	—	—	0	45.12	42.34	14.93
	BadNet	85.94	40.46	32.54	7.45	0	43.14	40.08	13.26
	TokenSwap	85.16	43.35	34.43	8.41	0	43.46	41.34	13.85

C.2 CROSS-DATASET COMPARISON

Extending the robustness contribution to additional model variants, we report cross-dataset results in Tables 6 and 7.

D MORE RESULTS OF ABLATION STUDIES

To show that our effectiveness contribution is robust to design choices, we run extensive ablations over poisoning rate, trigger type, trigger size, and trigger location, with results in Table 8. The headline is that across essentially all poisoning choices TokenSwap exceeds 80% ASR, the lone exception being the very low poisoning rate of 0.1. For the poisoning rate, a higher rate yields a higher ASR because more backdoor samples are available to learn from, though the ASR saturates once the rate exceeds 0.5. For the trigger type, TokenSwap reaches satisfactory ASRs with most trigger patterns,

Table 5: Attack performance on the MSCOCO dataset. Consistent with the other benchmarks, TokenSwap’s high attack success rate (ASR) on poisoned inputs demonstrates effectiveness, while its R-1 (Rouge-1), R-L (Rouge-L), and BLEU scores—comparable to the clean model on clean inputs—demonstrate preserved model utility. All metrics are reported in percentages (%).

Model	Attack Type	Poisoned Input (Attack Effectiveness)				Clean Input (Model Utility)			
		ASR ($\uparrow$)	R-1	R-L	BLEU	ASR	R-1	R-L	BLEU
BLIP2	Clean Model	—	—	—	—	0	43.15	38.22	8.27
	BadNet	39.06	40.04	34.79	6.10	0.78	42.11	37.40	7.24
	TokenSwap	75.00	40.04	31.60	4.79	0	41.49	37.11	7.80
InstructBlip	Clean Model	—	—	—	—	0	40.60	35.75	6.98
	BadNet	57.81	31.84	26.28	2.66	0	39.58	35.36	7.17
	TokenSwap	57.81	41.28	33.07	7.07	0	41.60	36.72	8.10
LLaVA-7B	Clean Model	—	—	—	—	0	40.48	36.23	8.55
	BadNet	67.97	38.72	30.51	4.63	0	39.75	34.70	8.82
	TokenSwap	74.22	39.28	29.51	3.10	0	42.07	36.87	9.42
LLaVA-13B	Clean Model	—	—	—	—	0	40.57	36.26	8.52
	BadNet	64.84	37.46	28.94	1.94	0	41.15	35.59	8.73
	TokenSwap	77.34	39.59	30.09	4.24	0	41.61	36.39	9.69

Table 6: Cross-dataset evaluation on LLaVA-7B of our TokenSwap against the Baseline. The attacked model is fine-tuned on the poisoned *MSCOCO* and evaluated on *Flickr8k* and *Flickr30k*. As on BLIP-2, TokenSwap suffers a smaller ASR drop under domain shift, reinforcing the robustness contribution.

Attack Type	Evaluation Setting	Poisoned Input (Attack Effectiveness)				Clean Input (Model Utility)			
		ASR ($\uparrow$)	R-1	R-L	BLEU	ASR	R-1	R-L	BLEU
BadNet	<i>in Flickr8k</i>	83.59	38.87	36.20	11.09	0	45.89	43.31	14.29
	MSCOCO→Flickr8k	68.75 (-14.84)	37.84	31.10	4.76	0	39.29	36.42	7.76
TokenSwap	<i>in Flickr8k</i>	89.06	39.75	31.58	5.75	0	42.21	39.42	12.30
	MSCOCO→Flickr8k	80.47 (-8.59)	38.47	30.90	4.56	0	40.60	36.79	7.62
BadNet	<i>in Flickr30k</i>	78.91	35.85	28.43	6.06	0	40.37	34.55	10.28
	MSCOCO→Flickr30k	63.28 (-15.13)	31.41	25.21	2.48	0	34.49	29.59	3.25
TokenSwap	<i>in Flickr30k</i>	85.16	37.49	29.02	6.10	0	40.07	33.93	10.30
	MSCOCO→Flickr30k	74.22 (-10.94)	31.76	25.04	2.57	0	33.53	29.41	3.81

Table 7: Cross-dataset evaluation on LLaVA-7B of our TokenSwap against the Baseline (grouped by evaluation setting). The attacked model is fine-tuned on the poisoned *MSCOCO* and evaluated on *Flickr8k* and *Flickr30k*, again showing TokenSwap’s stronger generalization under domain shift.

Attack Type	Evaluation Setting	Poisoned Input (Attack Effectiveness)				Clean Input (Model Utility)			
		ASR ($\uparrow$)	R-1	R-L	BLEU	ASR	R-1	R-L	BLEU
<i>in Flickr8k</i>	BadNet	83.59	38.87	36.20	11.09	0	45.89	43.31	14.29
	TokenSwap	68.75 (-14.84)	37.84	31.10	4.76	0	39.29	36.42	7.76
	BadNet	89.06	39.75	31.58	5.75	0	42.21	39.42	12.30
	TokenSwap	80.47 (-8.59)	38.47	30.90	4.56	0	40.60	36.79	7.62
MSCOCO→Flickr8k	BadNet	78.91	35.85	28.43	6.06	0	40.37	34.55	10.28
	TokenSwap	63.28 (-15.13)	31.41	25.21	2.48	0	34.49	29.59	3.25
	BadNet	85.16	37.49	29.02	6.10	0	40.07	33.93	10.30
	TokenSwap	74.22 (-10.94)	31.76	25.04	2.57	0	33.53	29.41	3.81

and it stays stable in attack effectiveness across trigger sizes and locations. On the utility side, the conclusion is equally clean: across all poisoning settings, TokenSwap does not meaningfully degrade the text quality of answers on clean inputs.

Table 8: Ablation study of TokenSwap over poisoning rate, trigger type, trigger size, and trigger location, evaluated on BLIP2 and Flickr8k. The consistently high ASR shows the effectiveness contribution is robust to these design choices.

Ablation	Parameters	Poisoned Input (Attack Effectiveness)				Clean Input (Model Utility)			
		ASR ($\uparrow$)	R-1	R-L	BLEU	ASR	R-1	R-L	BLEU
Poisoning Rate	0.1	75.00	44.7	35.17	9.86	0	45.23	42.27	12.86
	0.3	83.59	41.80	32.97	7.21	0	45.37	42.13	13.75
	0.5	91.41	44.06	34.67	9.10	0	45.51	41.67	12.79
	0.7	92.97	43.04	35.21	9.96	0	44.82	41.36	13.08
	0.9	92.97	43.22	35.21	9.96	0	44.86	41.25	13.08
Trigger Type	Black	87.50	42.69	33.21	7.88	0	45.37	41.91	12.54
	Blended (noise)	92.19	43.10	33.56	7.99	0	45.59	42.25	12.09
	Blended (hello kitty)	85.16	43.05	34.32	8.36	0	45.95	41.95	13.25
	SIG	81.25	42.06	33.14	7.78	0	45.89	41.49	12.34
	WaNet	92.19	42.89	33.56	7.37	0	45.32	42.14	12.74
Trigger Size	10	88.28	42.91	34.7	8.26	0	45.5	41.39	12.41
	20	89.84	42.14	32.85	7.36	0	45.56	41.95	12.60
	30	91.41	44.06	34.67	9.10	0	45.51	41.67	12.79
	50	90.63	42.06	33.27	7.36	0	45.19	41.61	12.83
Trigger Location	Left Top	92.19	42.60	32.72	6.60	0	45.16	41.78	12.62
	Four Corners	87.50	42.14	32.72	7.10	0	45.99	42.72	12.79
	Middle	92.19	42.48	33.99	8.34	0	45.94	42.39	13.25